Neural Networks

Lab 3 – Task 1

Penguin Classification

Perceptron and Adaline

Visualization and Analysis Report

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1 Aim

The aim of this task is to classify penguins using two algorithms: Perceptron and Adaline.

In each test, we choose two features and two penguin species. We train the model, draw a decision line, and check the results on test data.

We use five different combinations for each algorithm. This report includes both high and lower accuracy results.

2 Data and Preparation

The lab dataset has 150 samples: 50 Adelie, 50 Chinstrap, and 50 Gentoo.

The available features are CulmenLength, CulmenDepth, FlipperLength, OriginLocation, and BodyMass.

The data preparation steps are:

- Select two species and two features.
- Give the first species label $+1$ and the second species label -1.
- Shuffle each class using `random_state=42`.
- Use 30 samples from each class for training and the other 20 for testing.
- Fill missing numerical values with the training mean.
- Fill missing non-numerical values with the most common training value.
- Scale the selected features using the training mean and standard deviation.

Each test has 60 training samples and 40 test samples. No rows are removed to handle missing values.

OriginLocation is changed into numbers before training:

Location	Code
Torgersen	0
Biscoe	1
Dream	2

These numbers are category codes, not physical measurements.

3 Models and Test Settings

Both models use:

$$z = w_1 x_1 + w_2 x_2 + b.$$

The final class is $+1$ when $z \geq 0$, and -1 when $z < 0$. The decision line is:

$$w_1 x_1 + w_2 x_2 + b = 0.$$

The weights and bias start with small random values between -0.1 and 0.1. Both algorithms use seed 42 before this step. The seed keeps the same starting values when the same experiment is repeated. The starting values are still randomly generated, not set to zero.

Perceptron changes its weights when the predicted class is wrong. It can stop early if a full training epoch has no mistakes.

Adaline uses the linear output z to find the error. The cost used in our code is:

$$J = \frac{1}{2N} \sum_{i=1}^{N} (t_i - z_i)^2.$$

Adaline stops when this cost is below the threshold, or when it reaches the epoch limit.

Setting	Value
Learning rate	0.01
Maximum epochs	50
Adaline cost threshold	0.01
Bias	Enabled
Initialization seed	42 for both algorithms
Training samples	60
Test samples	40

Table 1: Settings used in all comparisons.

The epoch value is a limit. It is not the exact number of epochs used in every run.

4 Evaluation

We calculate the confusion matrix manually. We do not use `scikit-learn.metrics` or another ready-made confusion matrix function.

The first species is the positive class:

- TP: a first-species sample predicted correctly.
- FN: a first-species sample predicted as the second species.
- FP: a second-species sample predicted as the first species.
- TN: a second-species sample predicted correctly.

Accuracy is:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \times 100\%.$$

The plots show training samples and the decision line. The accuracy and confusion matrix use test samples. The plot axes show scaled feature values.

5 Comparison 1: CulmenLength and CulmenDepth

Classes: Adelie (+1) and Chinstrap (−1).

Model	Accuracy	TP	FP	FN	TN
Perceptron	100%	20	0	0	20
Adaline	97.5%	20	1	0	19

Table 2: Comparison 1 test results.

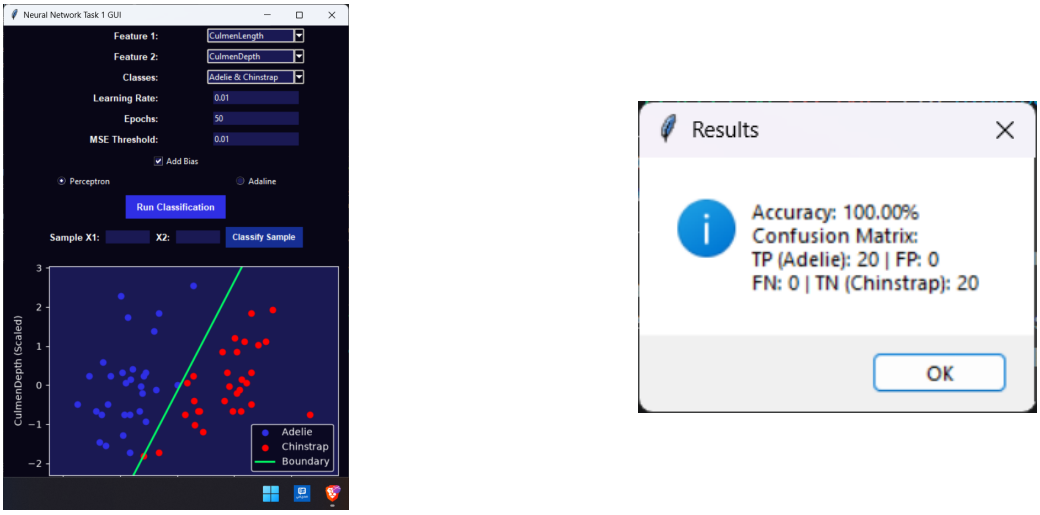


Figure 1: P1: Perceptron using CulmenLength and CulmenDepth for Adelie and Chinstrap.

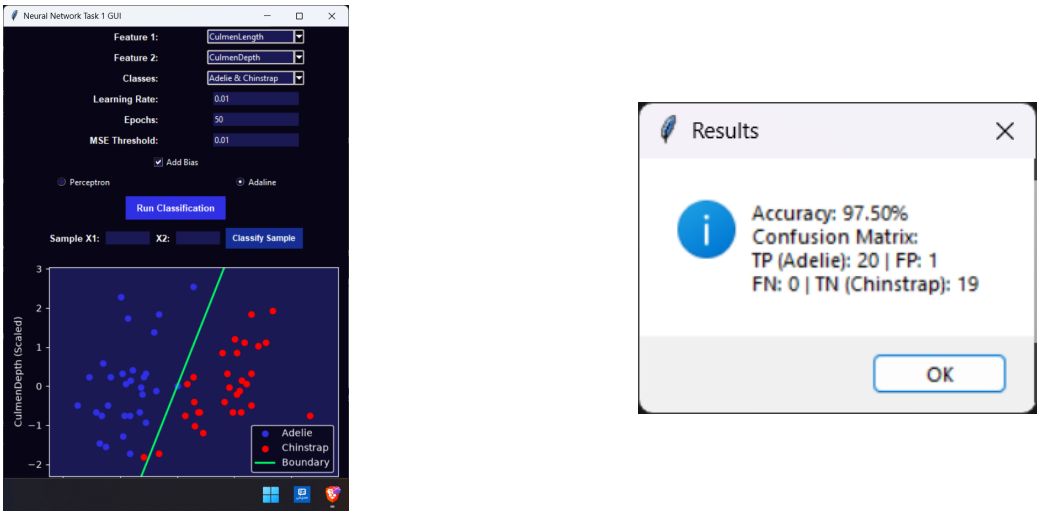


Figure 2: A1: Adaline using CulmenLength and CulmenDepth for Adelie and Chinstrap.

Analysis

Perceptron classified all 40 test samples correctly. Adaline classified 39 samples correctly. It predicted one Chinstrap sample as Adelie. These features worked well for this class pair. Bill length and bill depth helped both models tell the two species apart in this test. Perceptron had one more correct prediction than Adaline. A score of 100% here does not mean that all future samples will be correct.

6 Comparison 2: CulmenDepth and BodyMass

Classes: Chinstrap (+1) and Gentoo (−1).

Model	Accuracy	TP	FP	FN	TN
Perceptron	100%	20	0	0	20
Adaline	100%	20	0	0	20

Table 3: Comparison 2 test results.

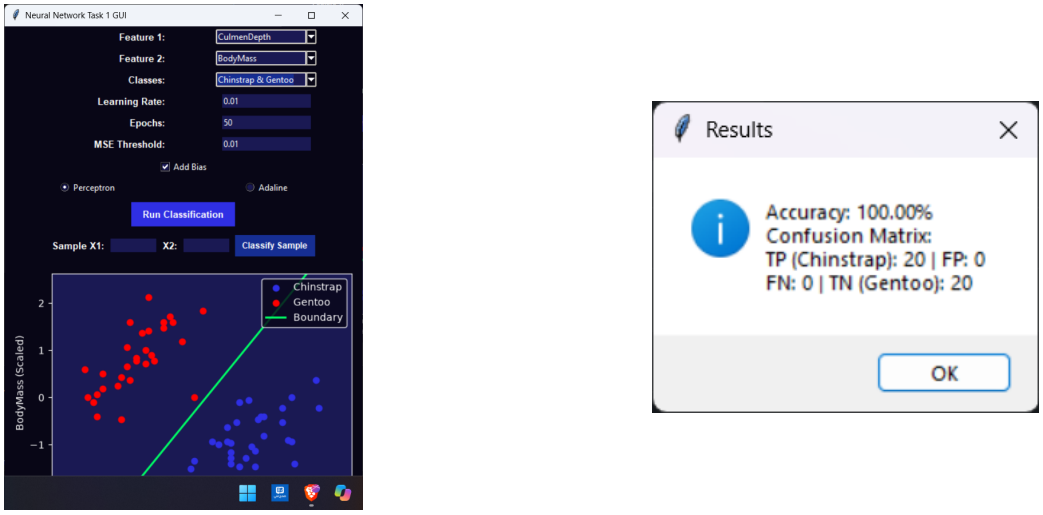


Figure 3: P2: Perceptron using CulmenDepth and BodyMass for Chinstrap and Gentoo.

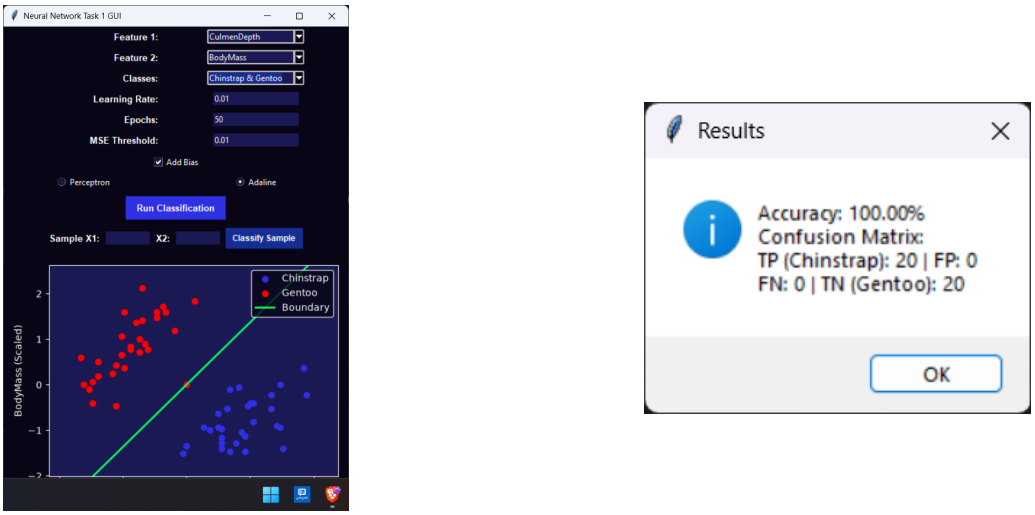


Figure 4: A2: Adaline using CulmenDepth and BodyMass for Chinstrap and Gentoo.

Analysis

Both models classified all 40 test samples correctly. There were no false positives or false negatives. CulmenDepth and BodyMass worked very well for this class pair in the recorded tests. This was the only combination where both models reached 100%. The same accuracy does not mean the models have the same decision line. Different lines can give the same answers for these test samples.

7 Comparison 3: CulmenLength and FlipperLength

Classes: Adelie (+1) and Chinstrap (−1).

Model	Accuracy	TP	FP	FN	TN
Perceptron	95%	19	1	1	19
Adaline	92.5%	19	2	1	18

Table 4: Comparison 3 test results.

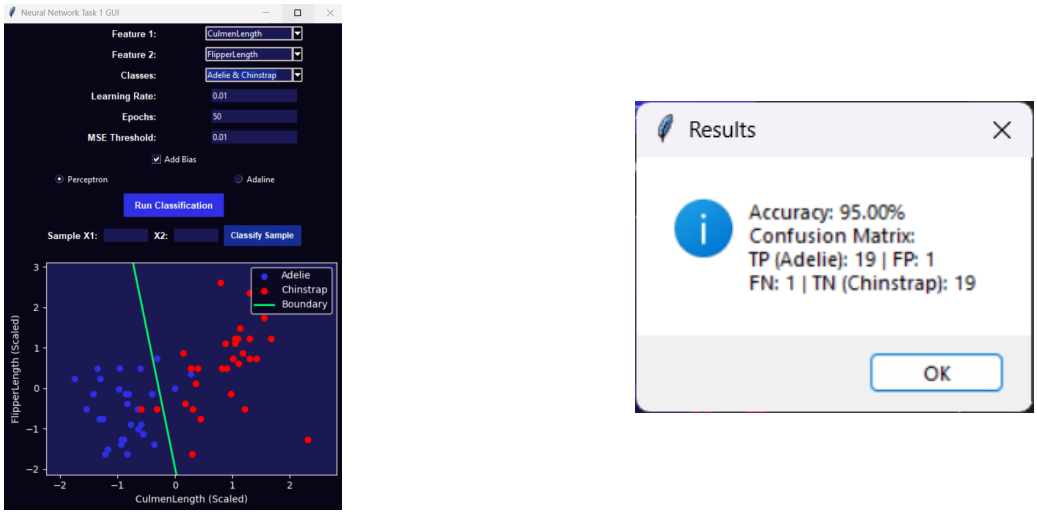


Figure 5: P3: Perceptron using CulmenLength and FlipperLength for Adelie and Chinstrap.

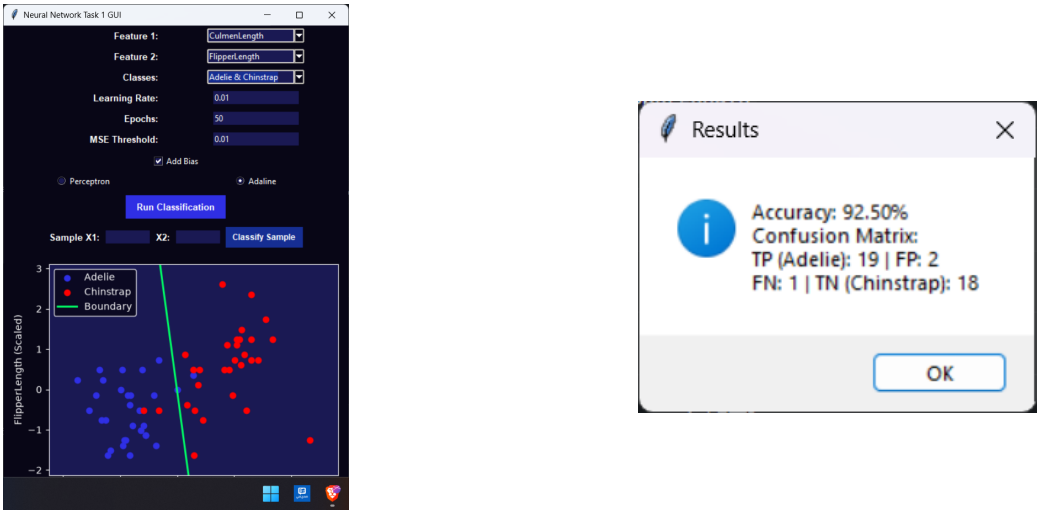


Figure 6: A3: Adaline using CulmenLength and FlipperLength for Adelie and Chinstrap.

Analysis

Perceptron classified 36 samples correctly and made two mistakes in each class. Adaline classified 37 samples correctly. It predicted two Chinstrap samples as Adelie and one Adelie sample as Chinstrap. Adaline had one more correct prediction than Perceptron. Both models had lower accuracy than in comparison 1. For this class pair, CulmenLength with CulmenDepth gave better results than CulmenLength with FlipperLength under the tested settings.

8 Comparison 4: OriginLocation and CulmenLength

Classes: Adelie (+1) and Gentoo (−1).

Model	Accuracy	TP	FP	FN	TN
Perceptron	90%	18	2	2	18
Adaline	95%	19	1	1	19

Table 5: Comparison 4 test results.



Figure 7: P4: Perceptron using OriginLocation and CulmenLength for Adelie and Gentoo.

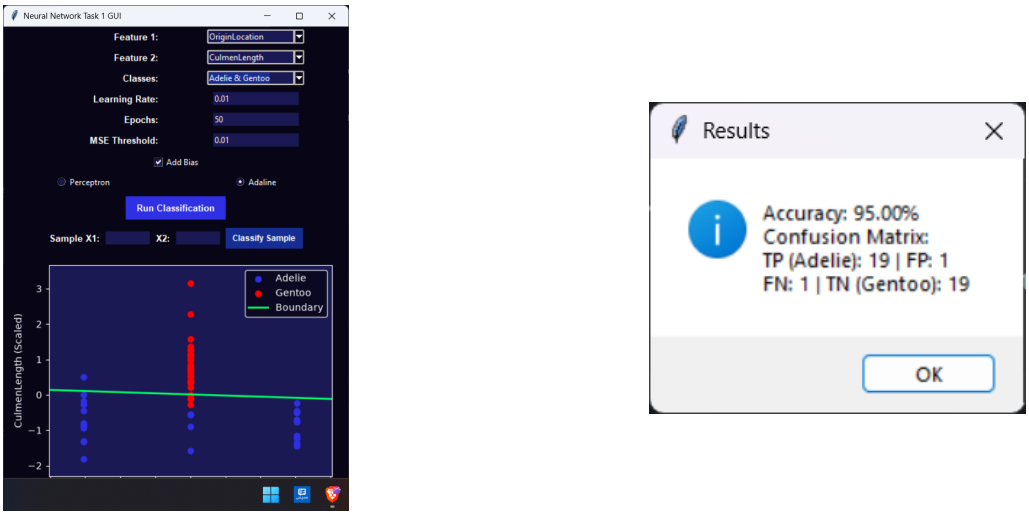


Figure 8: A4: Adaline using OriginLocation and CulmenLength for Adelie and Gentoo.

Analysis

Both models classified 38 samples correctly. Each model predicted one Adelie sample as Gentoo and one Gentoo sample as Adelie. They had the same accuracy and the same confusion matrix counts. These counts do not tell us whether the mistakes were on the same samples. OriginLocation and CulmenLength gave good results together. Location has only three possible codes, so its plot axis has only a few possible positions after scaling. We should read these as categories, not physical distances between places.

9 Comparison 5: FlipperLength and BodyMass

Classes: Adelie (+1) and Chinstrap (−1).

Model	Accuracy	TP	FP	FN	TN
Perceptron	70%	12	4	8	16
Adaline	85%	16	2	4	18

Table 6: Comparison 5 test results.

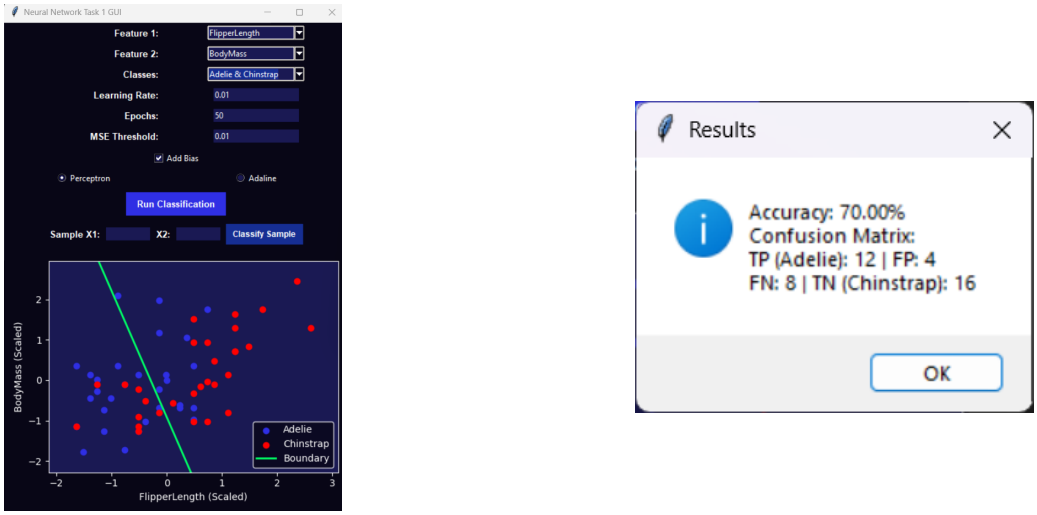


Figure 9: P5: Perceptron using FlipperLength and BodyMass for Adelie and Chinstrap.

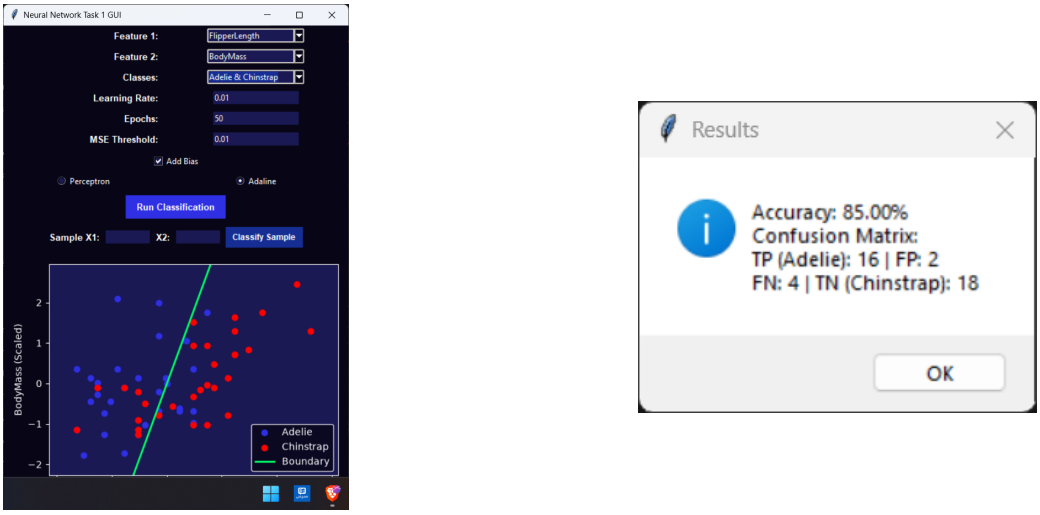


Figure 10: A5: Adaline using FlipperLength and BodyMass for Adelie and Chinstrap.

Analysis

This combination gave the lowest accuracy for both models in our tests. Perceptron classified 28 samples correctly and made 12 mistakes. Eight Adelie samples were predicted as Chinstrap, and four Chinstrap samples were predicted as Adelie. Adaline classified 34 samples correctly and made six mistakes. Four Adelie samples were predicted as Chinstrap, and two Chinstrap samples were predicted as Adelie. Adaline made fewer errors in both classes. It had six more correct predictions than Perceptron. These features were less useful for this class pair than the feature pairs in comparisons 1 and 3. This is our lower-performance example. The test results do not prove that a perfect line is impossible.

10 Summary and Conclusion

Comparison	Perceptron	Adaline
1	100%	97.5%
2	100%	100%
3	95%	92.5%
4	90%	95%
5	70%	85%

Table 7: Accuracy in the five comparisons.

Perceptron had higher accuracy in comparison 1. Both models had the same accuracy in comparisons 2 and 4. Adaline had higher accuracy in comparisons 3 and 5.

The highest Perceptron accuracy was 100%, using:

- CulmenLength and CulmenDepth for Adelie and Chinstrap;
- CulmenDepth and BodyMass for Chinstrap and Gentoo.

The highest Adaline accuracy was 100%, using CulmenDepth and BodyMass for Chinstrap and Gentoo.

CulmenDepth and BodyMass therefore gave the highest accuracy for both models in the same comparison.

For Adelie and Chinstrap, CulmenLength and CulmenDepth gave the best results among the tested feature pairs. FlipperLength and BodyMass gave the lowest results.

Perceptron accuracy ranged from 70% to 100%. Adaline accuracy ranged from 85% to 100%.

These tests show that feature choice is important. Neither model was better in every comparison.

Limits of the Results

Each test set has only 40 samples. One mistake changes accuracy by 2.5 percentage points.

Both models use seed 42 before weight and bias initialization. With the same data, settings, code, and software environment, this allows the same experiment to give the same result again.

However, we tested only one initialization seed. We did not compare results across different seeds, so these scores do not show how the models perform with every possible starting point.

Our conclusions apply to the tested combinations. They do not mean that one model or feature pair is always the best.